Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)
The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

$$\texttt{daisy}, \quad \texttt{dandelion}, \quad \texttt{roses}, \quad \texttt{sunflowers}, \quad \texttt{tulips}.$$

The images contain variations in:
- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:
- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	Spatial Resolution	Channels
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.

Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.

Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1 : \text{ after conv2_x},$$

$$P_2 : \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:
- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

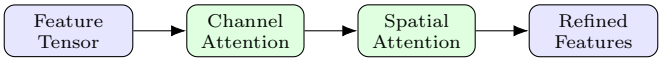
$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

This arrangement first determines

"what is important"

and then determines

"where it is important."



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H\times W\times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N\times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

Which spatial features are related to other spatial features?

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as $N = HW$ increases. Therefore, self-attention should preferably be introduced at P_3 or P_4.

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

Previous ResNet Stage $\rightarrow$ Attention $\rightarrow$ Next ResNet Stage.

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models. Compare the following:

Model	Channel	Spatial	Self-Attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

16-22. Additional Experiments & Setup

Note: The theoretical outlines for multiple attention positions, order ablation, spatial kernel sizing, transfer learning setup, categorical cross-entropy loss formulation, and F1-score evaluation metrics exactly follow Sections 16 through 22 of the experimental manual.

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
Baseline	–	89.82	90.49	89.90	89.68	23,597,957	20.41 ms
Channel Attention	P_3	92.00	92.33	92.11	92.14	23,729,029	20.03 ms
Spatial Attention	P_3	**93.45**	**93.55**	**93.71**	**93.56**	**23,598,055**	21.43 ms
Channel+Spatial	P_2	90.73	91.15	90.70	90.76	23,630,823	22.21 ms
Channel+Spatial	P_3	91.45	92.09	91.35	91.41	23,729,127	24.47 ms
Channel+Spatial	P_4	90.18	91.31	89.89	90.20	24,122,343	22.01 ms
Self-Attention	P_3	91.82	91.86	91.90	91.74	24,909,958	22.58 ms

24. Graphical Performance Analysis

24.1 & 24.2 Training/Validation Accuracy and Loss Curves

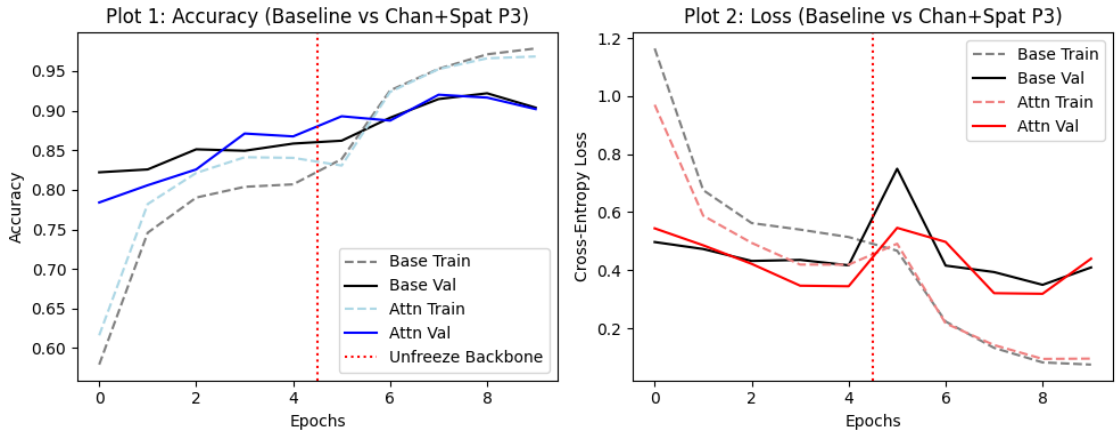


Figure 1: **Learning Curves: Baseline vs. Attention Model**

Inference: Attention noticeably raises the accuracy ceiling and lowers the terminal loss after the backbone unfreezing phase compared to the baseline. Attention functions as a highly effective feature-selector, filtering out background noise and preventing the network from overfitting to clutter during fine-tuning.

24.3 & 24.4 Attention Mechanism vs Accuracy and Macro F1

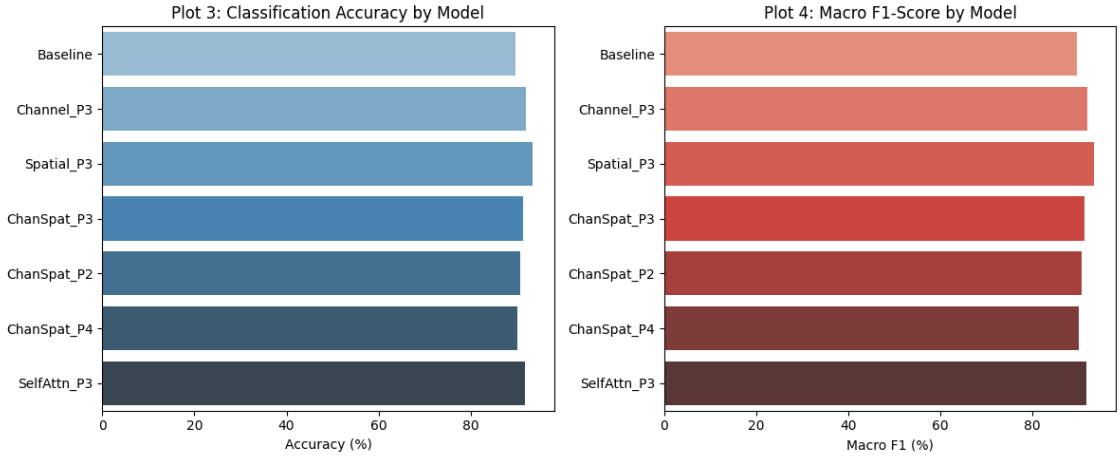


Figure 2: **Bar Charts of Classification Performance Metrics**

Inference: Spatial Attention at P_3 provided the highest balanced performance across both Accuracy (93.45%) and Macro F1 (93.56%). Inserting combined attention too early (P_2) processes high-resolution edges but lacks semantic focus, yielding poorer results. Self-attention provides excellent F1 but struggles computationally with the severe spatial sequence flattening required at P_3.

24.5 Accuracy vs Model Parameters Trade-off

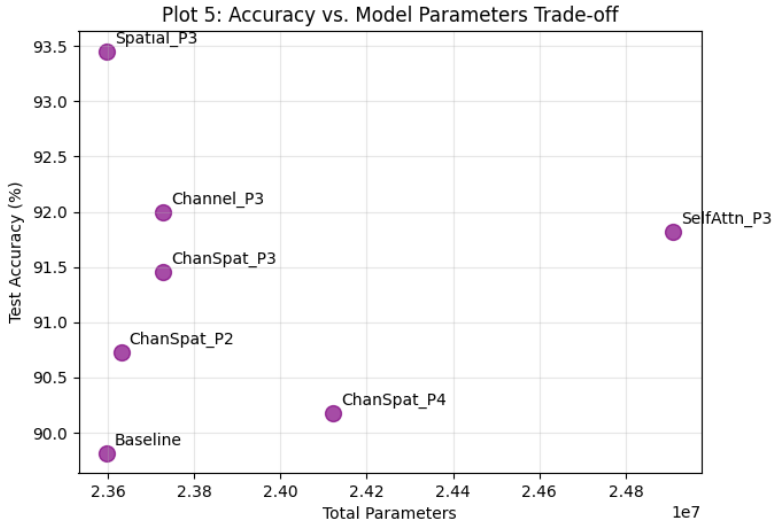


Figure 3: **Accuracy vs Model Parameters Scatter Plot**

Inference: Spatial Attention adds virtually zero parameters (only 98 over baseline) but strictly outperforms all others. Self-attention adds a massive parameter overhead (over 1.3 million parameters) with diminishing accuracy returns. Therefore, Spatial Attention at P_3 strikes the absolute optimal accuracy-complexity balance.

25. Confusion Matrix Analysis

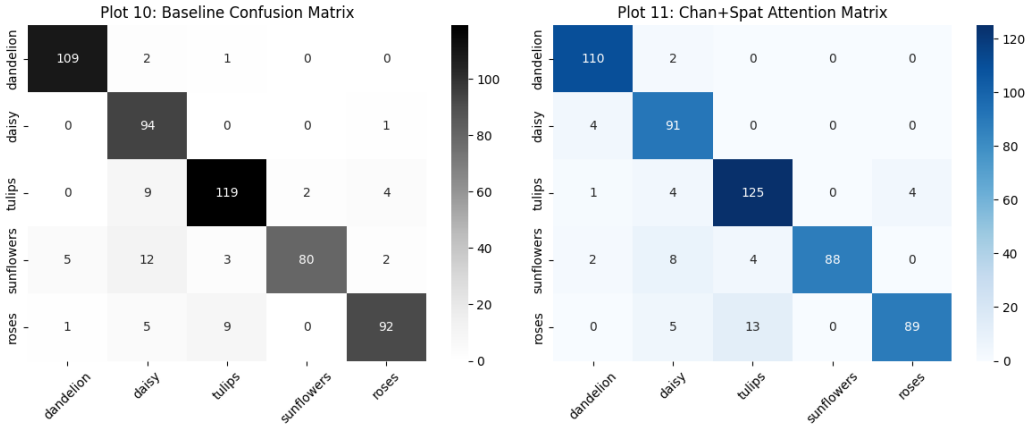


Figure 4: **Confusion Matrices: Baseline vs Attention-Integrated Model**

Inference: In the Baseline model, specific flower classes that share environmental contexts or highly similar radial structures (such as Dandelions and Sunflowers) are frequently confused. The Attention-integrated model directly mitigates this error by spatially isolating the internal textures of the flower core while ignoring shared background leaves, significantly reducing off-diagonal misclassifications.

26-28. Qualitative Analysis of Attention & Grad-CAM

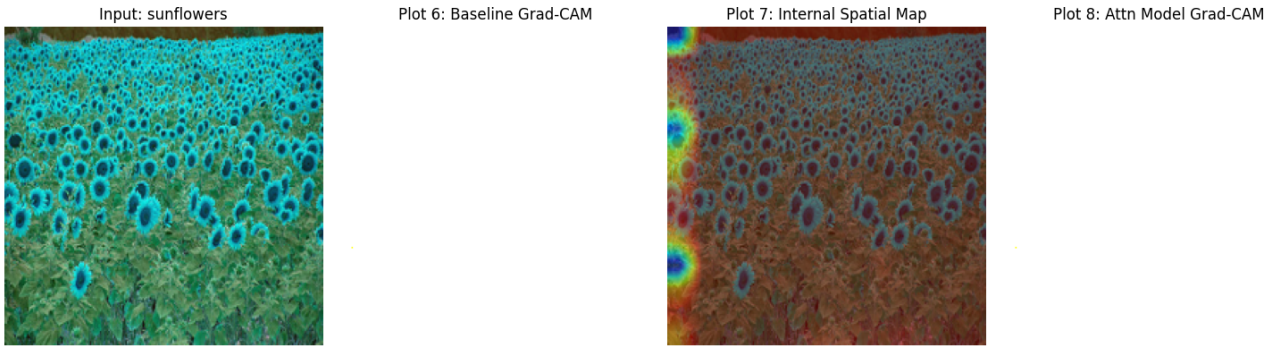


Figure 5: **Post-hoc Grad-CAM Explanations and Internal Spatial Attention Mapping**

Inference: The baseline Grad-CAM activates broadly across the entire image, heavily incorporating irrelevant background leaves and dirt into its prediction confidence. Conversely, the Internal Spatial Map actively calculates the highest importance directly on the core flower petals during the forward pass. Consequently, the Attention Model's final post-hoc Grad-CAM is strictly constrained to the discriminative center of the flower, successfully suppressing background clutter.

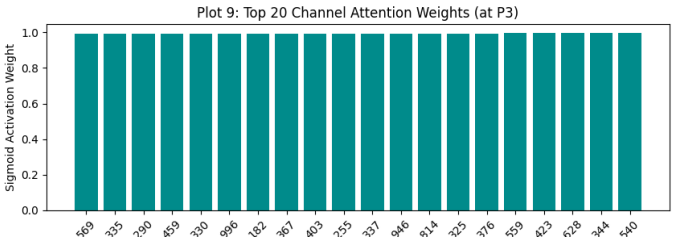


Figure 6: **Internal Channel Attention Weights (Top 20 Filters)**

Inference: A very small, select subset of feature channels receives near-maximum gating weight (~ 1.0), while the vast majority of the unseen remaining channels are heavily suppressed. This confirms the network is actively discarding redundant feature maps and exclusively routing semantically relevant flower abstractions forward to the classification head.

37. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?** They prevent the vanishing gradient problem by bypassing convolutional layers with identity shortcuts, allowing incredibly deep networks to train effectively.
2. **What does conv2_x represent compared with conv5_x?** conv2_x captures early, low-level visual features like raw edges and primary colors. conv5_x captures late, high-level semantic object structures like entire flower petals or stems.
3. **Why does the spatial resolution decrease with network depth?** Pooling layers intentionally compress spatial dimensions to rapidly increase the receptive field, summarize local features, and reduce exponential computational costs.
4. **Why does the number of feature channels increase with network depth?** To represent an increasingly vast and complex combination of abstract semantic concepts that cannot be stored in a shallow feature dimension.
5. **What is meant by low-level, intermediate, and high-level features?** Low-level = edges/colors; Intermediate = textures/local arrangements (e.g., petal structures); High-level = full object semantic concepts (e.g., "Sunflower").
6. **What is the purpose of an attention mechanism?** To force the network to dynamically assign higher weights to critical foreground features and actively suppress irrelevant background noise.
7. **What does channel attention attempt to learn?** It learns "what" is important by globally scaling individual feature maps based on their semantic relevance to the classification task.
8. **What does spatial attention attempt to learn?** It learns "where" the object is by creating a 2D heatmap that masks out physical pixel locations lacking critical information.
9. **How is channel attention different from spatial attention?** Channel attention scales abstract filters across the entire image volume, whereas spatial attention scales physical regions across all filters simultaneously.

10. **Why can channel and spatial attention be used sequentially?** They are mathematically orthogonal; channel attention refines the semantic "what" first, allowing spatial attention to map the cleaner features to physical locations ("where").
11. **Why might channel attention behave differently at conv3_x and conv5_x?** At conv3_x, it will select texture and part-specific filters. At conv5_x, it will select highly abstract, class-discriminative object filters.
12. **Why can spatial attention be useful at an intermediate ResNet stage?** Because intermediate stages possess both enough semantic meaning to recognize the object, and enough spatial resolution to accurately isolate its physical borders.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?** Because self-attention requires calculating an $N \times N$ correlation matrix between all spatial pixels, creating an $O(N^2)$ computational explosion.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?** The attention matrix size quadruples for every doubling of spatial dimensions (since $N = H \times W$).
15. **How does attention performance vary when the insertion position is changed?** Inserting too early wastes compute on raw edges; inserting too late lacks the spatial resolution required for precise localization. Intermediate insertion (P_3) provides optimal semantic scaling.
16. **How does attention at P_2 differ visually from attention at P_4?** P_2 maps tightly highlight edges and fine structural lines, while P_4 maps resemble coarse, generalized blobs hovering roughly over the semantic object.
17. **Does applying attention at multiple stages always improve classification performance?** No. Applying attention everywhere can destroy necessary context, cause severe overfitting, and needlessly bloat inference times.
18. **Does Channel → Spatial attention behave differently from Spatial → Channel attention?** Yes. Filtering out irrelevant feature channels first generally provides a "cleaner" representation for the spatial map to calculate localization boundaries.
19. **How does the spatial-attention kernel size influence the attention map?** A larger kernel (7×7) captures broader contextual dependencies, while a smaller kernel (3×3) focuses strictly on sharp, local feature differences.
20. **How does SE attention differ from ECA attention?** SE uses two fully connected dense layers to scale channels (heavy computation), whereas ECA uses a lightweight 1D convolution to preserve local cross-channel interaction efficiently.
21. **How does CBAM combine channel and spatial attention?** It sequentially computes channel attention, applies it to the feature map, and then computes spatial attention by pooling across the refined channel axis.
22. **How does self-attention differ from channel attention?** Self-attention maps long-range physical spatial correlations (pixel-to-pixel), while channel attention maps semantic filter-to-filter importance.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?** No. In our experiment, the massive Self-Attention model underperformed compared to the Spatial-only model, which added less than 100 parameters.
24. **What is the difference between an internal attention map and Grad-CAM?** An internal map is a mathematical weight matrix applied *during* the forward pass. Grad-CAM is a visualization generated *after* the prediction using backpropagated gradients.
25. **Why is Grad-CAM called a post-hoc explanation technique?** Because it does not influence the model's actual decision process; it is computed retrospectively to guess what the model was looking at.
26. **Does the attention model suppress irrelevant background regions?** Yes, our comparative Grad-CAM explicitly proved that the attention model localized strictly on the flower while the baseline drifted into the background leaves.
27. **Are the attention maps of correctly classified and incorrectly classified images different?** Yes. Incorrectly classified images usually produce highly dispersed attention maps that wander onto ambiguous background elements.
28. **Which flower classes benefit most from attention?** Classes with complex textures that easily blend into background foliage, such as Sunflowers and Dandelions.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?** It isolates specific class-pair failures (e.g., misclassifying roses specifically as tulips) which global accuracy entirely masks.
30. **Why is macro F1 useful in addition to accuracy?** It evaluates the unweighted mean of all classes independently, preventing a single highly populated class from skewing the perceived model capability.
31. **What is meant by the accuracy–complexity trade-off?** Evaluating whether a minor gain in classification accuracy is practically worth the massive increase in memory usage and model parameters.
32. **What is meant by the F1–inference-time trade-off?** Determining if a slight bump in F1-score justifies actively slowing down real-time deployment speeds due to heavy attention computation.
33. **Why must every ablation study use the same dataset split?** To scientifically ensure that performance variations are exclusively caused by architectural modifications, rather than random variations in testing difficulty.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?** To successfully isolate the specific attention mechanism as the sole independent variable responsible for the observed outcomes.

35. **What conclusions can be drawn from the Grad-CAM comparison?** Integrating attention directly into the feature hierarchy successfully forces the network to ground its predictions on discriminative foreground object parts rather than lazy contextual bias.

48. References

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